

Aryan Sood

Curriculum Vitae

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EDUCATION

Stanford University | MS Mechanical Engineering (GPA: 3.83/4.0)

Stanford, CA

University of California, Berkeley | BS Mechanical Engineering (GPA: 3.73/4.0)

Berkeley, CA

TECHNICAL EXPERTISE

CAD/PLM: SOLIDWORKS (PDM Specialist), PTC Creo, Windchill, AutoCAD MEP, Revit, Fusion 360, Civil 3D, 3ds Max.

Analysis: FEA (Mesh Independence/Convergence), CFD (Airflow Simulation), Abaqus, ANSYS, MATLAB, Python (ROOT).

Manufacturing: Waterjet, CNC Machining, Carbon Fiber Composites, Vacuum Tech (TVAC), Vibration Testing, FMEA.

Product Realization: Failure Analysis, Ruggedized Systems, Manual Milling, Sand Casting, Mechanical Seals, Fastening Systems.

PROFESSIONAL EXPERIENCE

Carnegie Robotics | Mechanical Engineering Intern

Pittsburgh, PA

- Lead mechanical design and fabrication for ruggedized autonomous and robotic systems, specifically focused on vehicle platform upfits and material handling optimization.
- Engineered precision mounts, IP-rated enclosures, and rotary hydraulic systems featuring integrated mechanical seals and optimized fastening systems.
- Developed specialized tooling for robotic manipulation systems and managed assembly, testing, and structural validation through SOLIDWORKS PDM.
- Validated mechanical robustness of autonomous vehicle platforms through vibration testing and iterative prototyping using waterjet and CNC fabrication.

Virtual Presenz Inc. | Mechanical Engineering Intern

San Francisco Bay Area

- Designed and prototyped a rugged pneumatic lasso trigger mechanism for intruder incapacitation, incorporating a breakaway shell system.
- Engineered internal engaging hooks designed to latch onto clothing upon shell fragmentation, validated through iterative force-of-engagement testing.

CK Architecture Interiors | MEP Engineering Intern

Dubai, UAE

- Executed fluid mechanics and heat transfer calculations for complex residential and corporate HVAC duct systems to ensure thermal efficiency.
- Optimized HVAC layouts and equipment placement in AutoCAD MEP, conducting insulation and efficiency analysis for medium-sized residential buildings.
- Developed plumbing systems in AutoCAD MEP featuring flow rate and stress analysis; simulated flow conditions and system efficiency in Autodesk Revit.

Moving Universe Lab (UC Berkeley) | ME Research Apprentice

Berkeley, CA

- Developed five conceptual designs for deployable space telescope baffles for the CuRIOS CubeSat mission under Astronomy Chair Dr. Jessica Lu.
- Engineered and modeled three fully deployable, compactly packed mechanisms utilizing springs and motors for autonomous deployment in orbital environments.
- Conducted comprehensive FEA with stress-strain testing and mesh independence studies to ensure structural rigidity under high-G launch loads.
- Executed environmental qualification testing, including mechanical and thermal vacuum (TVAC) cycles, on designed CubeSat mechanical interfacing and electronic payloads.

Alef Aeronautics | Mechanical Engineering Intern

San Mateo, CA

- Responsible for the design, modeling, and manufacturing of fully functional, flight-capable eVTOL automotive prototypes.
- Managed end-to-end carbon fiber and composite polymer manufacturing, including complex layering, resin preparation, application, and finishing.
- Conducted static and dynamic testing of suspension systems to identify mechanical failure modes; executed sheet metal manufacturing for structural chassis components.
- Overseas a high-output 3D printer fleet, performing assembly, installation, and 24/7 operation for rapid iterative assembly of aerodynamic show models.

Space Technologies at California (STAC) | Mechanical Engineer

Berkeley, CA

- Executed 3D modeling and Factor of Safety (FoS) analysis of a specialized helium tank transportation and storage apparatus.
- Performed dynamic testing of servo-actuated parafoil descent systems to investigate drag and steering parameters for autonomous post-launch recovery.
- Designed and manufactured ruggedized payload covers to protect sensitive cutoff mechanisms and parafoil attachments during descent.

Redington Gulf | Mechanical Design Intern

Dubai, UAE

- Utilized PTC Creo to model and simulate complex snowmobile engine mechanics for performance and thermal optimization.
- Executed Product Lifecycle Management (PLM) workflows using Windchill to manage engineering data and design iterations.

Undergraduate Lab @ Berkeley | Student Researcher

Berkeley, CA

- Authored a technical research paper on CERN LHC theoretical principles, focusing on particle kinematics and energetics.
- Simulated high-energy particle collisions using a Python-based ROOT program to perform large-scale data analysis and verification.

PROJECTS

ME310: Luxury Hotel Experience Design (Stanford x Venture Singapore)

In Progress

- Collaborating with Venture Singapore to redefine luxury hospitality; conducting needfinding to map guest and staff pain points and quantify luxury metrics.
- Currently building experience and functional hardware prototypes to validate new service interaction models and mechanical hardware interventions.

NH35A Build & Bronze Sand-Cast Display Stand

Personal

- Built a regulated Seiko NH35A automatic watch featuring a "Bruce Wayne" aesthetic, oyster bracelet, and exhibition back.
- Manufactured a bespoke bronze display stand via wood pattern carving, green sand casting, and manual surface milling.

Algorithmic Guitar Stand & Soft Robotic Venus Flytrap

Academic

- Designed an FDM-printed guitar stand using Monolith algorithmic modeling to optimize organic geometry and material usage.
- Engineered a pneumatically actuated soft robot mimicking snapping mechanisms with autonomous sensing for contact-triggered closure.

TECHNICAL PUBLICATIONS

Hannah C. Gulick, Jessica Lu, Steven Beckwith, Kodi Rider, Jared Haertel, Charles-Antoine Claveau, Josh Bloom, Wei Liu, Dan Werthimer, Matt Dexter, **Aryan Sood**, et al. "Space-qualification of a commercial camera package with the Sony IMX455 CMOS detector for the CuRIOS-ED." *SPIE UV/Optical/IR Space Telescopes XII*, Sep 2025.

Hannah C. Gulick, Jessica R. Lu, **Aryan Sood**, Steve VW Beckwith, Joshua S. Bloom, Kodi Rider, Dan Werthimer, Wei Liu, Guy Nir, Harrison Lee, Jeremy McCauley. "CuRIOS-ED: the technology demonstrator for the CubeSats for rapid infrared and optical surveys mission." *SPIE Space Telescopes and Instrumentation*, Sep 2024.

Hannah Gulick, Jessica Lu, **Aryan Sood**, Steven Beckwith, Charles-Antoine Claveau, Joshua Bloom, Kodi Rider, Dan Werthimer, Wei Liu, Guy Nir, Harrison Lee, Jeremy McCauley. "Advantages of a Space-Based, All-sky Survey for Time-Domain Astrophysics." *AAS Meeting Abstracts*, Jan 2025.

VOLUNTEERING

Beti Padhao Foundation | Individual Contributor

India

- Delivered seminars and courses on Health Awareness and English literacy in remote villages to encourage public education enrollment.

Mohammed bin Rashid Center for Special Education | Student Volunteer

Abu Dhabi

- Supported children and adolescents diagnosed with Autism Spectrum Disorder through educational and recreational assistance.